TamNetClub – System Design Overview

1. Problem Statement

Existing social platforms prioritise open-ended discussion, where individual stories and dilemmas are often lost within broader conversations, noise, or unrelated engagement.
Users seeking perspective or judgment on personal situations frequently receive limited visibility, inconsistent responses, or no meaningful resolution. Additionally, comment-driven environments tend to fragment attention, shifting focus away from the original context toward general debate or unrelated contributions.
This creates a gap between:
users who want to share structured, personal dilemmas
and systems designed primarily for unstructured interaction and content engagement
TamNet$^{\text{Club}}$ addresses this gap by providing a dedicated environment for:

- story centrism
- Structured, comparable responses
- outcomes are resolved through consensus
- and consistent, outcome driven interaction.

TL;DR

TamNet$^{\text{Club}}$ is a story-centric, verdict-driven social platform that enforces structured interaction. Members submit cases, vote using constrained verdicts, and build standing through a ranking system based on:

- participation
- consensus accuracy
- consistency
- moderation signals

The system is designed to be moderation-first, high-signal, and difficult to game, with product constraints directly shaping the architecture.

2. Product Model

TamNet$^{\text{Club}}$ replaces open discussion with structured judgment:
Core Principles

- Gated membership – access is controlled
- Structured participation – no free-form comments
- Explicit moderation – enforcement is built-in
- Prestige-driven ranking – credibility is earned

Core Actions
Members can:

- apply to join / re-enter using nickname + PIN
- submit real-world cases
- vote on cases using limited verdicts
- report content and share cases
- view profiles, case history, and ranking progress

Product Rationale
The system is designed to shift social interaction from:

- open discussion → structured judgment

That product decision influences nearly every engineering choice in the system:

- story-centric
- no comment system
- tightly scoped verdicts
- explicit case windows
- moderation-first
- ranking system

3. High-Level Architecture

TamNet[Club] follows a layered full-stack architecture:
Client → API → Services → Repositories → Database

Stack

- Frontend: React 19, Vite, React Router
- Backend: Spring Boot, Spring Security, REST API
- Database: Relational model with Flyway-managed migrations
- Security: Spring Security + custom session filter
- Analytics: PostHog

Architectural Responsibilities

Frontend

- handles routing and protected views
- manages onboarding, reentry UX, voting, profile views, reporting flows

Backend API

- owns all domain logic

Service Layer

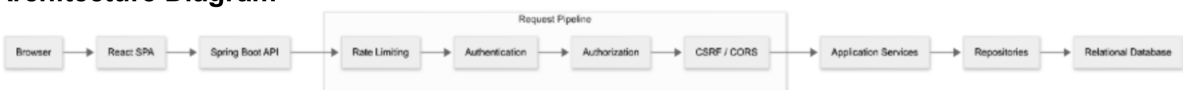
- centralises business rules

Repository Layer

- abstracts database access
- provides domain-specific queries

4. System Architecture

Architecture Diagram



System Architecture – Request flow from client through security pipeline to domain services and persistence layer.

Key Design Insight
Security is part of the request lifecycle, not a separate layer.
This ensures:

- consistent enforcement
- reduced attack surface
- predictable behaviour

5. Core Flows

Join / Approval Flow
Join application → review → approved/rejected
Reentry Flow
Reentry → authenticate → session issued
Voting Flow
Vote → validate → store → update case
Ranking Flow
Ranking → compute signals → update standing

6. Authentication & Security Design

TamNet^{Club} uses dual authentication models.
Member Auth
Member requests are authenticated through a dedicated session cookie.
Admin Auth
Admin login is handled through Spring Security.
Security Controls

- CSRF
- CORS
- Rate limiting
- Session validation

7. Ranking System Design

Signals: Activity, consensus, recency and risk
Ranking is multi-factor and designed to resist manipulation.

8. Product Trade-offs

No Comments Initially

- reduces moderation complexity
- keeps interaction constrained and analysable

Verdict-Based Interaction

- easier to moderate
- easier to reason about
- directly compatible with ranking and consensus logic

72-Hour Voting Window

- encourages timely engagement
- simplifies resolution rules
- prevents endless re-litigation of older cases

Prestige Over Speed
The rank system is intentionally strict. The product favours earned credibility over fast progression.

Summary

TamNet$^{\text{Club}}$ demonstrates:

- full-stack engineering across React, Spring Boot, and a relational data layer
- security awareness through session design, CSRF, CORS, rate limiting, and auth separation
- product thinking through constrained interaction design, moderation workflows, and gated membership
- system design through ranking logic, verdict resolution, recovery models, and beta operational planning

The system is designed as a product-oriented architecture, where domain logic, clear service boundaries, and security considerations are treated as first-class concerns.